

5.4 Conics Summary Worksheet

1. Find the vertex of the graph $y = 3x^2 - 12x + 7$.
2. Consider the equation of a conic: $x^2 + 2x + y^2 + 8y - 8 = 0$.
 - a. Does this equation represent a parabola, ellipse, circle, or hyperbola?
 - b. Carefully sketch a graph of the above equation.
3. Consider the equation of a conic: $-x^2 - 8y - 4 = 4y^2 - 4x$
 - a. Does this equation represent a parabola, ellipse, circle, or hyperbola?
 - b. Carefully sketch a graph of the above equation.
4. Consider the equation of a conic: $x^2 - 4y^2 - 2x + 8y - 7 = 0$
 - a. Does this equation represent a parabola, ellipse, circle, or hyperbola?
 - b. Carefully sketch a graph of the above equation.
5. Consider the equation of a conic: $x^2 = 6y - y^2$
 - a. Does this equation represent a parabola, ellipse, circle, or hyperbola?
 - b. Carefully sketch a graph of the above equation.
6. Consider the equation: $y^2 - 8x - 8y - 8 = 0$
 - a. Does this equation represent a parabola, ellipse, or hyperbola?
 - b. Carefully sketch a graph of the above equation.
7. Consider the conic given by the equation: $-x^2 + 2x + 4y^2 + 8y - 8 = 0$.
 - a. Does this equation represent a parabola, ellipse, or hyperbola?
 - b. Carefully sketch a graph of the above equation.